

Jenkins vs GitHub Actions Beginner Guide and Cheatsheet

Choosing between Jenkins and GitHub Actions depends on whether you want complete infrastructure control or instant managed automation. This guide breaks down the essential setup steps and commands for both tools so you can start automating builds today.

What you'll learn:

- 1 Setting up your first GitHub Actions workflow
- 2 Running Jenkins locally using Docker
- 3 Testing GitHub Actions locally with act
- 4 Writing a basic Declarative Jenkinsfile

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Setting up your first GitHub Actions workflow

GitHub Actions lives directly inside your repository with zero server setup required. You simply create a hidden folder named `.github/workflows` and add a single YAML configuration file.

This file tells GitHub what actions to take whenever events like a git push happen.

In this simple example, GitHub spins up an Ubuntu virtual machine and echoes a success message.

```
name: Quick Start
on: [push]
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - run: echo 'Pipeline running successfully!'
```

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Running Jenkins locally using Docker

Jenkins requires a server host, but you can run it locally on your laptop using Docker. This command downloads the official Jenkins Long Term Support (LTS) image and starts the container.

Port 8080 grants access to the Jenkins web UI, while port 50000 connects build agents. Once running, visit <http://localhost:8080> in your web browser to complete initial setup.

```
$docker run -d --name my-jenkins -p 8080:8080 -p 50000:50000 jenkins/jenkins:lts
```

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Testing GitHub Actions locally with act

Pushing tiny commit changes just to debug a broken GitHub Actions workflow is frustrating.

The open-source tool named 'act' uses Docker to run your GitHub Actions workflows locally.

It reads your .github/workflows folder and simulates the exact GitHub runner environment. Run this single command in your terminal root directory to test your workflows locally.

```
$act -j test
```

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Writing a basic Declarative Jenkinsfile

Jenkins pipelines are defined using a declarative code format inside a text file named Jenkinsfile.

This file defines agents where jobs execute and breaks automation into distinct readable stages.

Each stage contains step-by-step shell commands that execute sequentially.

Commit this file to your repository root so Jenkins can read and execute your pipeline automatically.

```
$pipeline {  
  agent any  
  stages {  
    stage('Build') {  
      steps {  
        sh 'echo Building Application...'  
      }  
    }  
  }  
}
```

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Key Takeaways

- + Use GitHub Actions for fast setup and small to mid-sized projects.
- + Use Jenkins if you require complex on-premises security or legacy tool integrations.
- + Always store build logic inside repository files (Jenkinsfile or GitHub YAML) rather than UI forms.
- + Test workflows locally using Docker or CLI tools before pushing to production.
- + Keep pipeline steps small and isolated so build failures are easy to pinpoint.

Watch Out For

- ! Hardcoding credentials or API keys directly inside pipeline scripts instead of secret managers.
- ! Forgetting to limit build concurrency, leading to runaway cloud server costs or high CPU loads.
- ! Creating huge, monolithic build scripts instead of breaking tasks into reusable stages.
- ! Overcomplicating Jenkins server setup when GitHub Actions managed runners are sufficient.

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